Consider the following objective functions,

1. $f_1(x) = x^2$.
2. $f_2(x) = 1/|x|$.
3. $f_3(x) = 1/x$.

For each function, provide its infimum on $\mathbb{R}$, or show that it does not exist. Provide also its minimum on $\mathbb{R}$, or show that it does not exist.

Answer the same question if the decision variable is constrained as follows:

$$-1 \leq x \leq 2.$$